

(12) **United States Patent**
Kominami et al.

(10) **Patent No.:** **US 12,397,983 B2**
(45) **Date of Patent:** **Aug. 26, 2025**

(54) **CLOSURE DEVICE AND SOFT PACKAGING CONTAINER**

(71) Applicant: **Toyo Seikan Co., Ltd.**, Tokyo (JP)

(72) Inventors: **Atsushi Kominami**, Kanagawa (JP);
Souji Araki, Kanagawa (JP); **Hiroki Tanaka**, Kanagawa (JP)

(73) Assignee: **Toyo Seikan Co., Ltd.**, Tokyo (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 125 days.

(21) Appl. No.: **17/800,894**

(22) PCT Filed: **Feb. 16, 2021**

(86) PCT No.: **PCT/JP2021/005777**

§ 371 (c)(1),

(2) Date: **Aug. 18, 2022**

(87) PCT Pub. No.: **WO2021/172113**

PCT Pub. Date: **Sep. 2, 2021**

(65) **Prior Publication Data**

US 2023/0086124 A1 Mar. 23, 2023

(30) **Foreign Application Priority Data**

Feb. 28, 2020 (JP) 2020-034292

(51) **Int. Cl.**

B65D 75/58 (2006.01)

B65B 51/04 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **B65D 83/771** (2025.01); **B65B 51/04** (2013.01); **B65B 51/10** (2013.01); **B65B 57/00** (2013.01);

(Continued)

(58) **Field of Classification Search**

CPC B65D 83/0055; B65D 75/5811;
B65D 31/16; B65D 75/566; B65B 51/04;
B65B 51/10; B65B 57/00

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,213,236 A 5/1993 Brown
6,269,837 B1 * 8/2001 Arent B67D 7/0205
251/149.6

(Continued)

FOREIGN PATENT DOCUMENTS

CN 104670706 A 6/2015
CN 105882973 A 8/2016

(Continued)

OTHER PUBLICATIONS

ISA/237) Written Opinion of the International Search Authority for International Patent Application No. PCT/JP2021/005777, issued/mailed by the Japan Patent Office on Apr. 13, 2021.

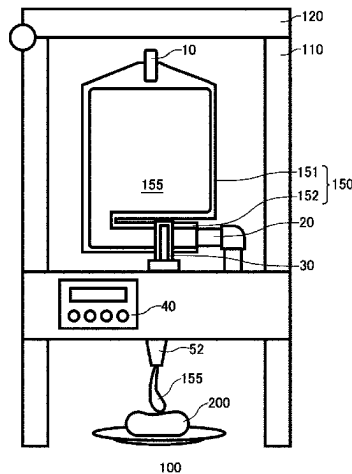
(Continued)

Primary Examiner — Frederick C Nicolas

(57) **ABSTRACT**

Provided is a closure device for closing a soft packaging container, the closure device including a holding section configured to hold a soft packaging container, a dispense section configured to dispense contents of the soft packaging container, a closure section configured to close the soft packaging container, and a control section configured to control timing for closing the soft packaging container by the closure section. The control section may include an acquisition section configured to acquire time information related to timing for closing the soft packaging container. Provided is a soft packaging container for accommodating contents, the soft packaging container including a main

(Continued)



region for accommodating the contents, and an extension region provided by being extended from the main region.

2016/0101430 A1 4/2016 Hoeft
2017/0043870 A1 2/2017 Wu
2019/0210726 A1 7/2019 Wu

16 Claims, 20 Drawing Sheets

FOREIGN PATENT DOCUMENTS

(51) Int. Cl.

B65B 51/10 (2006.01)
B65B 57/00 (2006.01)
B65D 30/10 (2006.01)
B65D 75/56 (2006.01)
B65D 83/00 (2006.01)

(52) U.S. Cl.

CPC **B65D 31/16** (2013.01); **B65D 75/566**
(2013.01); **B65D 75/5811** (2013.01)

CN	107826251	A	3/2018
CN	110816842	A	2/2020
EP	2332453	A1	6/2011
JP	H1084837	A	4/1998
JP	2010001007	A	1/2010
JP	2013224167	A	10/2013
JP	2017531492	A	10/2017
JP	2018516197	A	6/2018
JP	2019172355	A	10/2019
KR	20150003981	A	1/2015
TW	208691	B	7/1993
TW	200415081	A	8/2004

OTHER PUBLICATIONS

(56)

References Cited

U.S. PATENT DOCUMENTS

6,390,329	B1 *	5/2002	Maddox		A47K 5/1208	222/25
8,240,508	B2 *	8/2012	Wegelin		G05B 15/02	222/105
8,550,378	B2 *	10/2013	Mazooji		B05B 3/02	222/88
9,504,360	B2 *	11/2016	Van Diepen		A47K 5/1217	222/173
9,884,336	B2 *	2/2018	Hoeft		B05B 11/0054	222/173
10,500,699	B2	12/2019	Chang			
2005/0284888	A1	12/2005	Rhodenbaugh			
2009/0212072	A1 *	8/2009	Fenton		B05B 11/0044	222/173
2009/0314783	A1	12/2009	Losinski			
2011/0114669	A1 *	5/2011	Yang		A47K 5/16	222/639
2014/0103066	A1	4/2014	Geert Norbert			

ISA/237) Written Opinion of the International Search Authority for International Patent Application No. PCT/JP2021/005773, issued/mailed by the Japan Patent Office on May 11, 2021.

Office Action issued for related Japanese Application No. 2020-034282, transmitted from the Japanese Patent Office on Sep. 26, 2023 (drafted on Sep. 13, 2023).

Office Action issued for related Chinese Application 202180007368.9, issued by The State Intellectual Property Office of People's Republic of China on Aug. 24, 2024.

Office Action issued for counterpart Taiwanese Application 110106843, transmitted from the Taiwan Intellectual Property Office on Sep. 30, 2024 (issued on Sep. 25, 2024).

Office Action issued for related U.S. Appl. No. 17/801,267, issued by the US Patent and Trademark Office on Mar. 18, 2025.

Office Action issued for counterpart Chinese Application 202180015853.0, issued by The State Intellectual Property Office of People's Republic of China on Jun. 4, 2025.

* cited by examiner

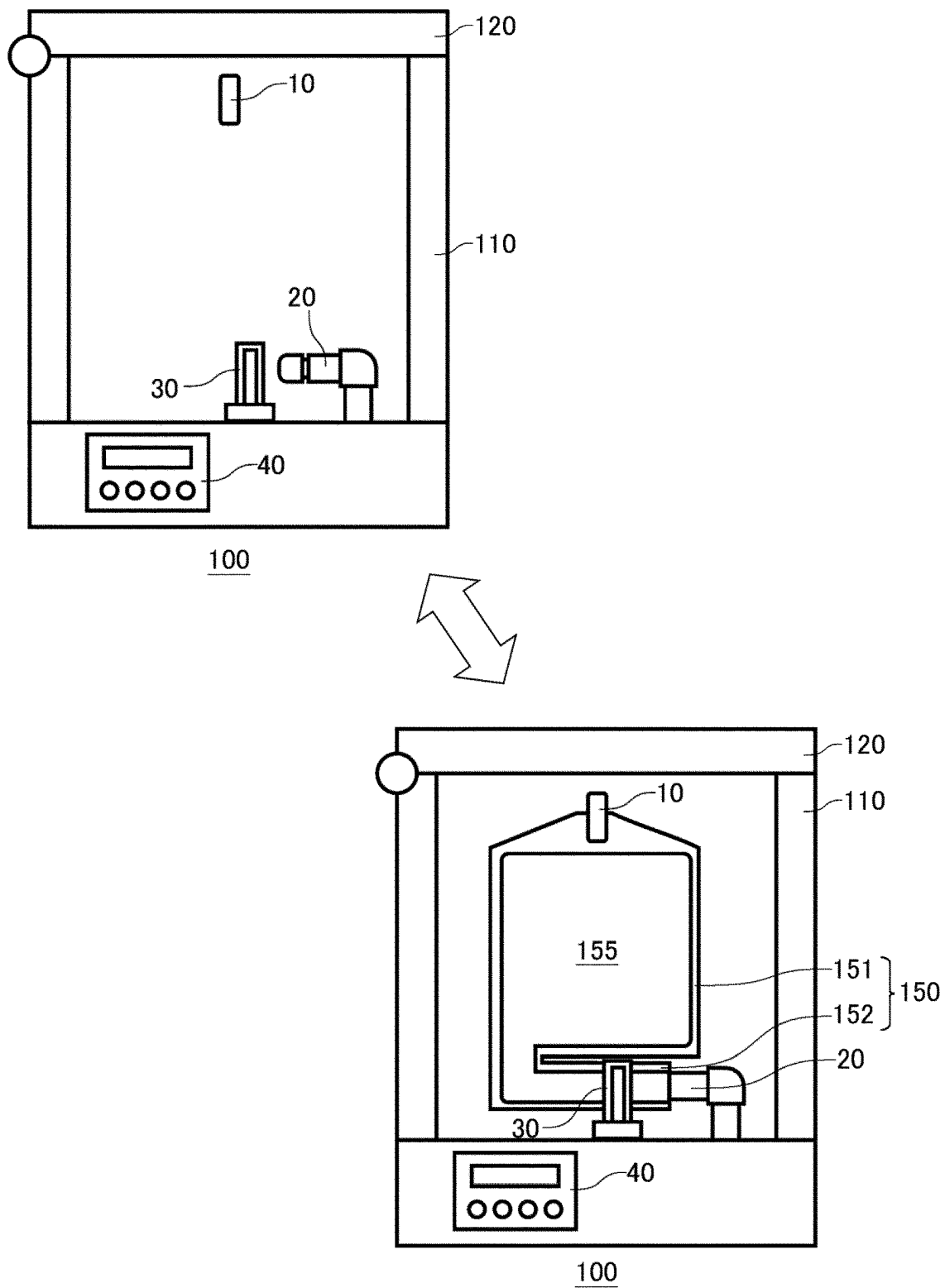


FIG. 1A

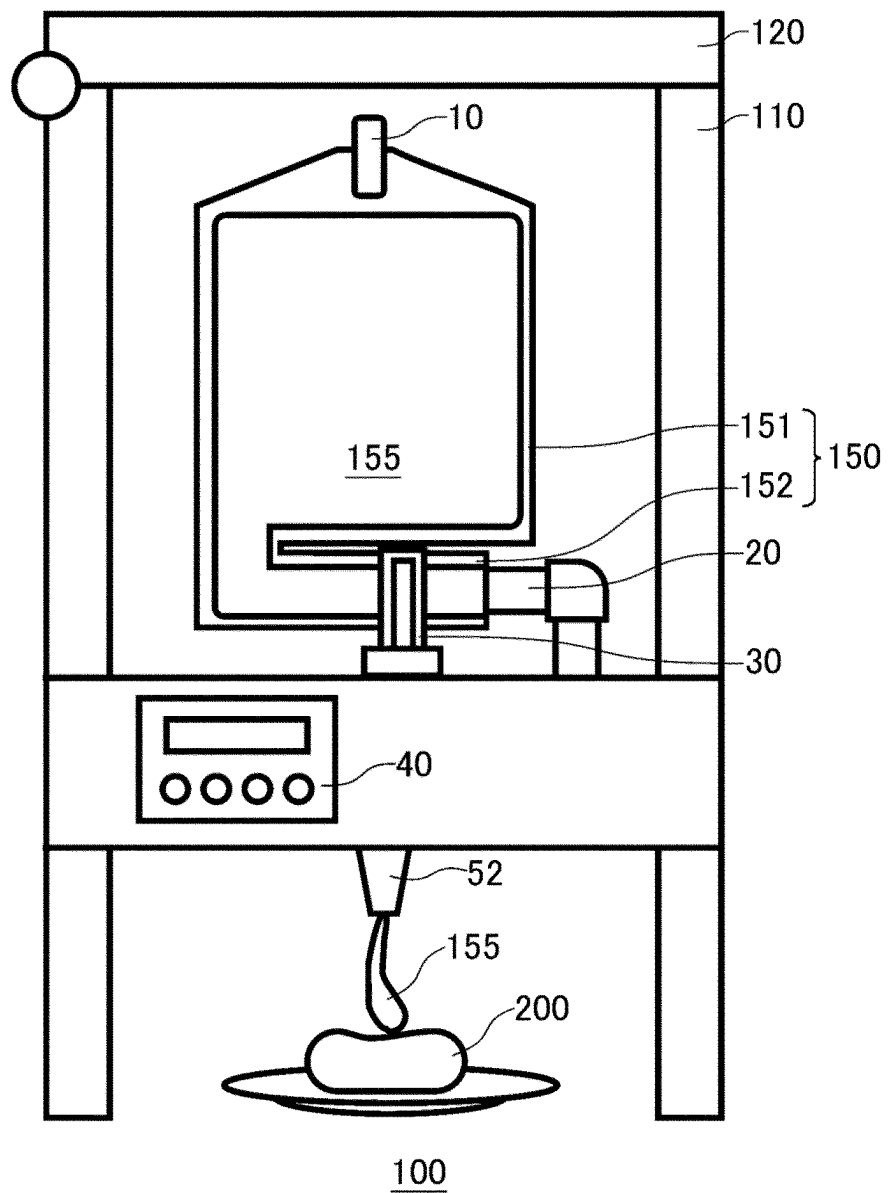


FIG. 1B

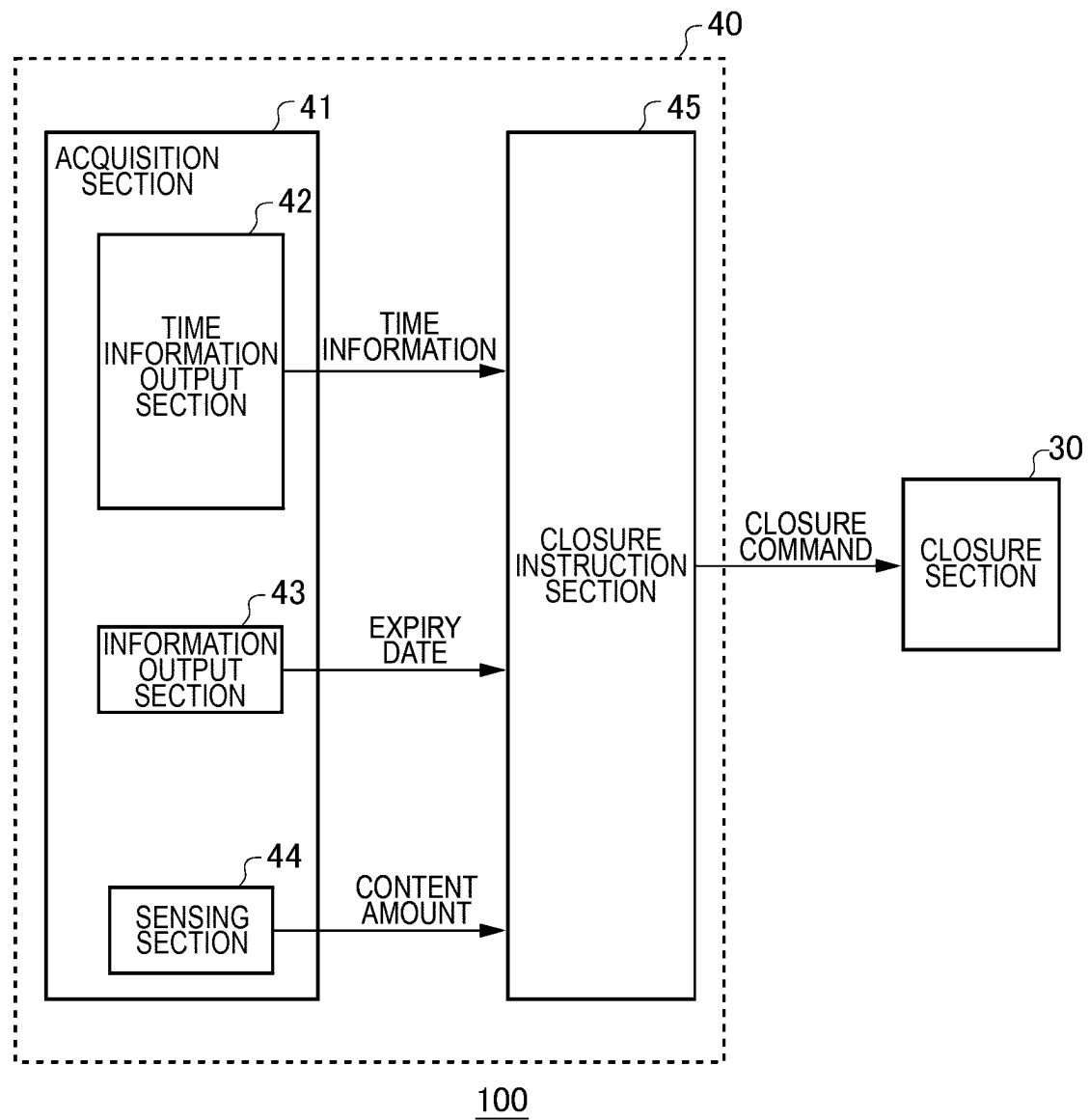
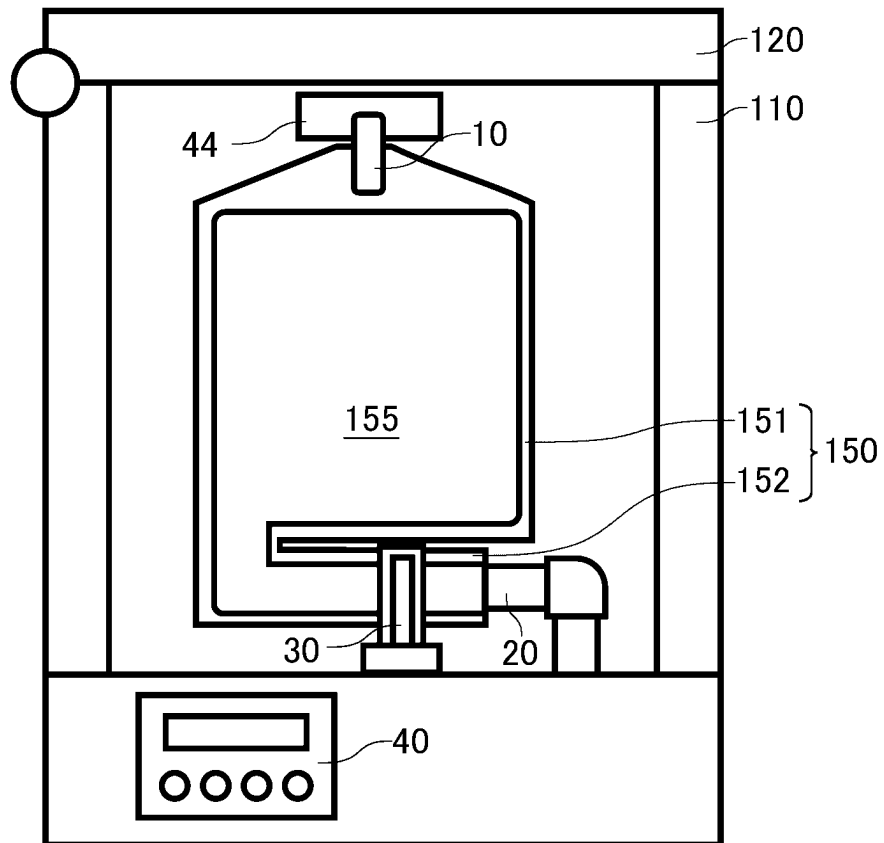


FIG. 2



100

FIG.3

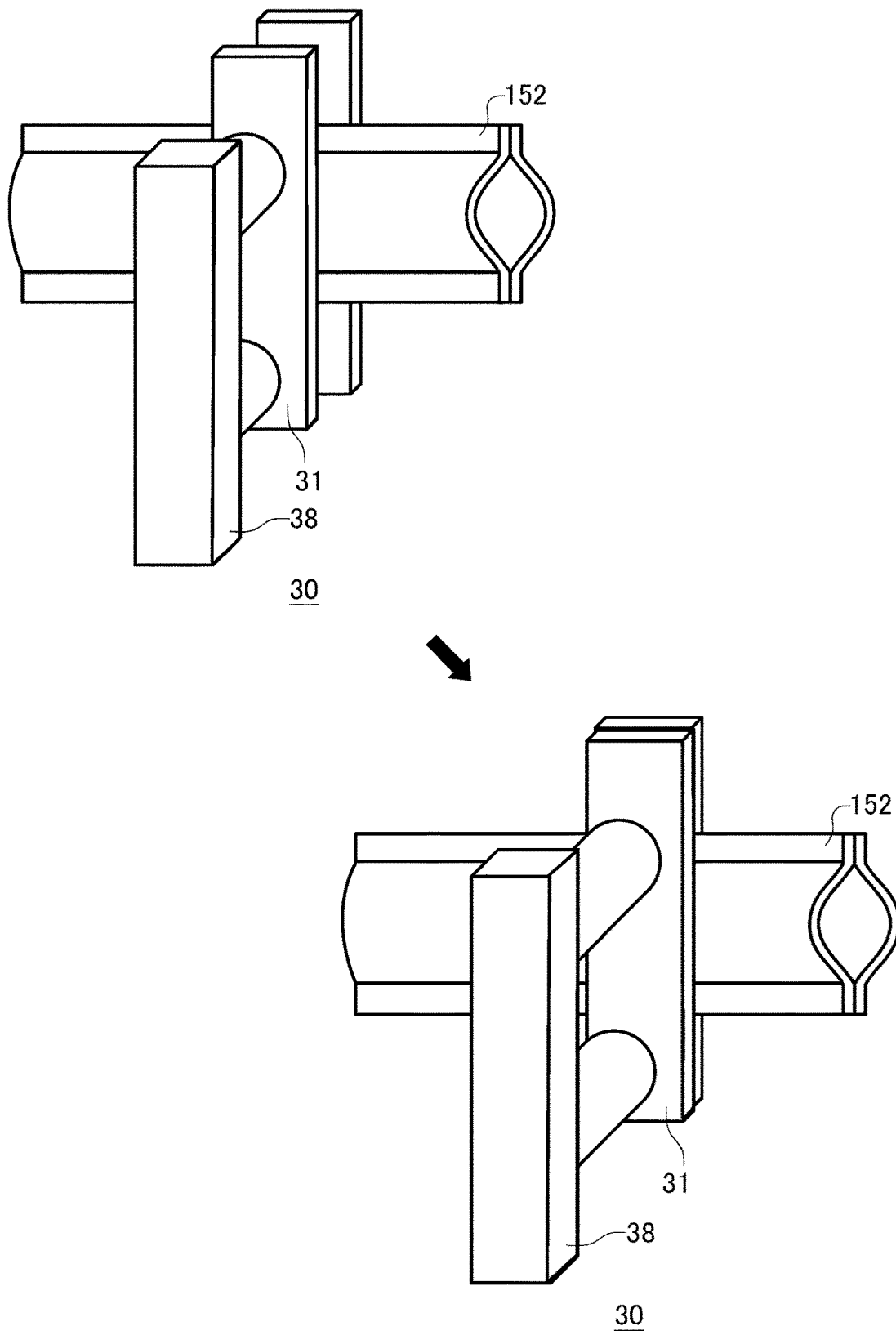


FIG. 4A

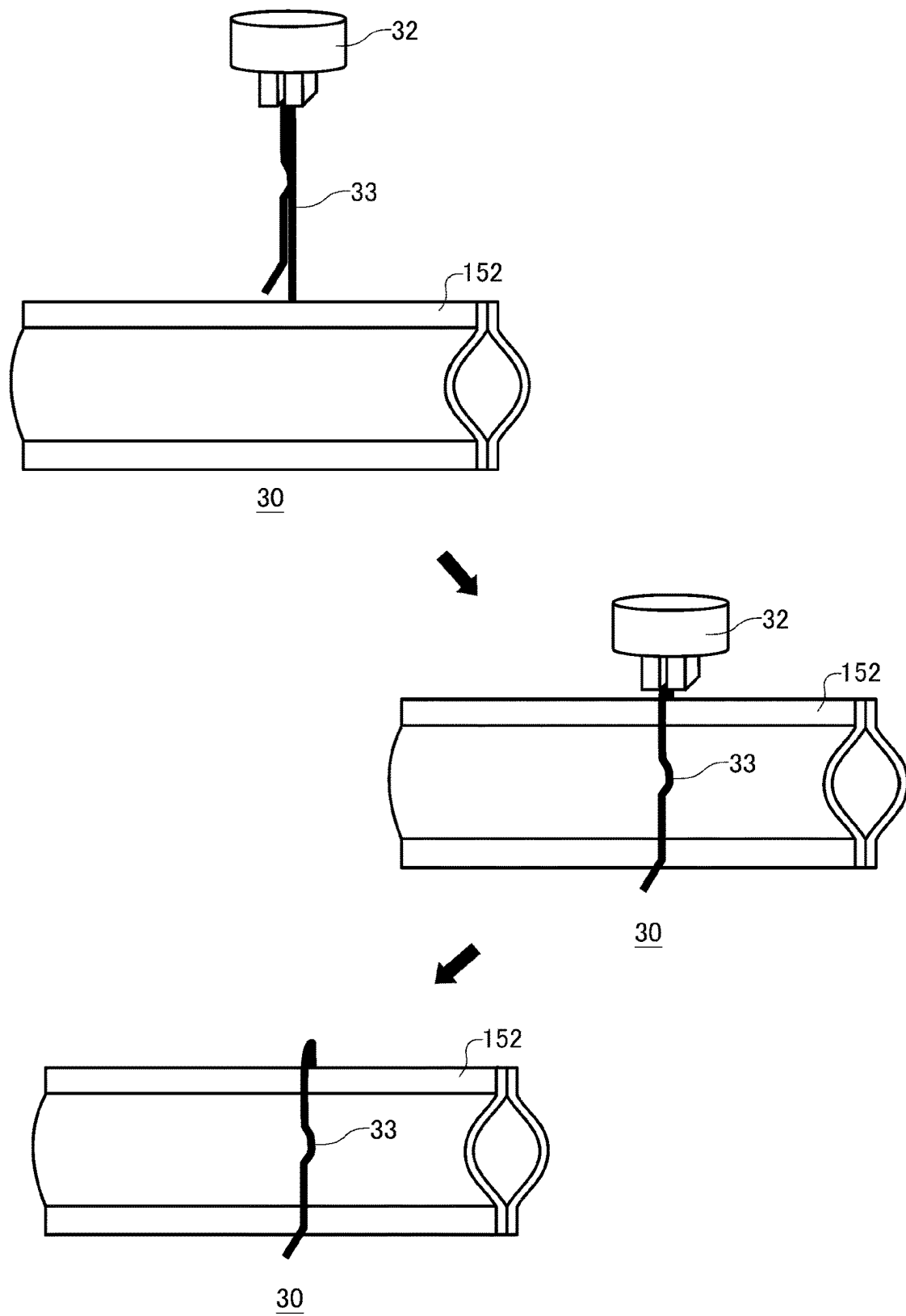


FIG. 4B

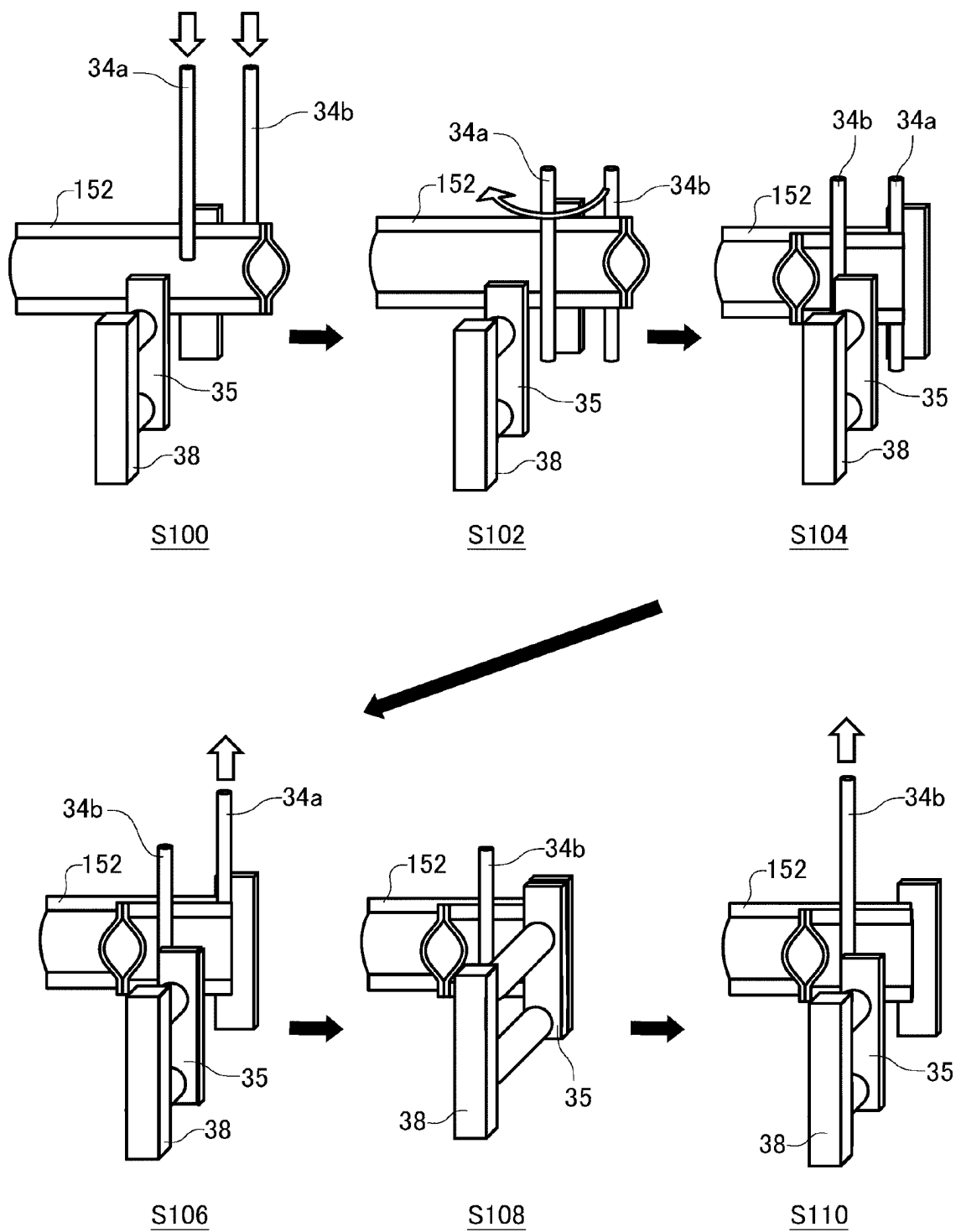


FIG. 4C

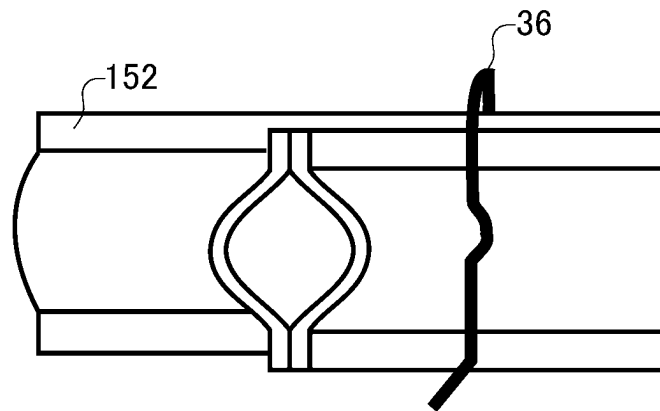


FIG.5A

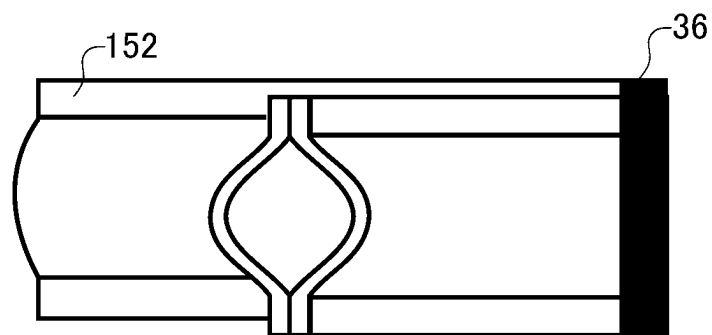


FIG. 5B

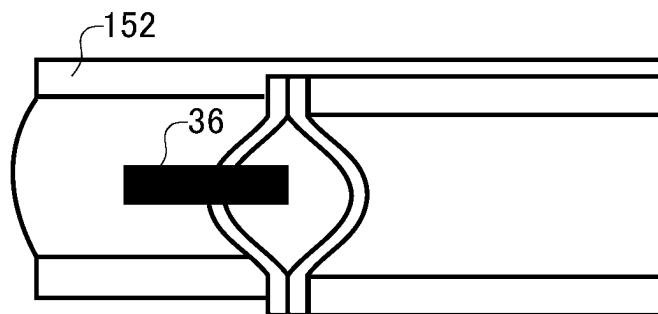


FIG.5C

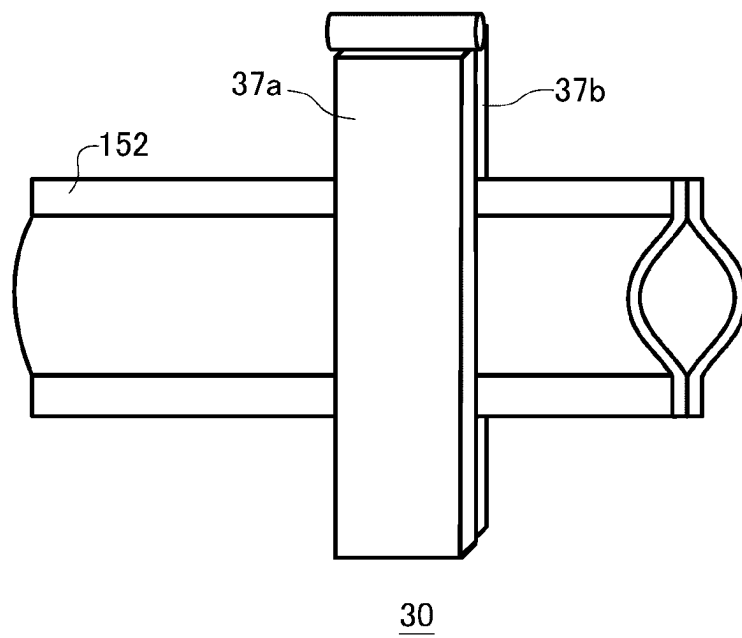
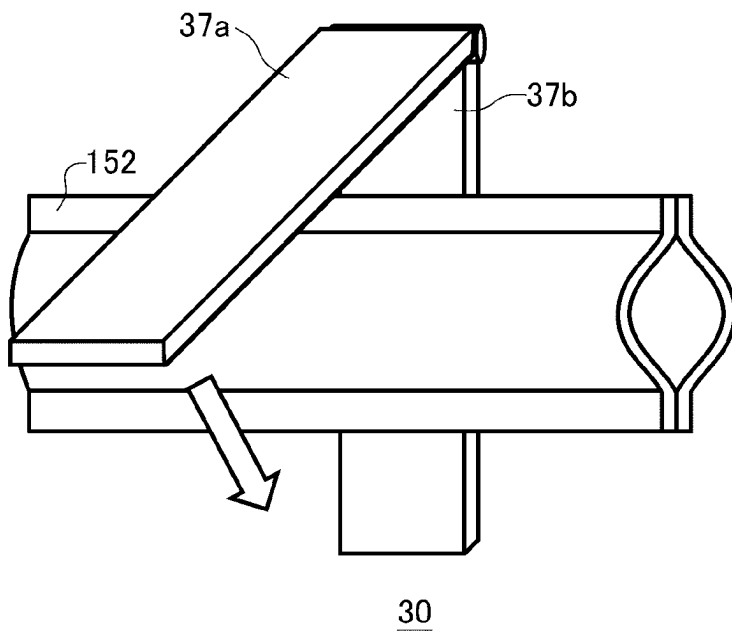


FIG. 6

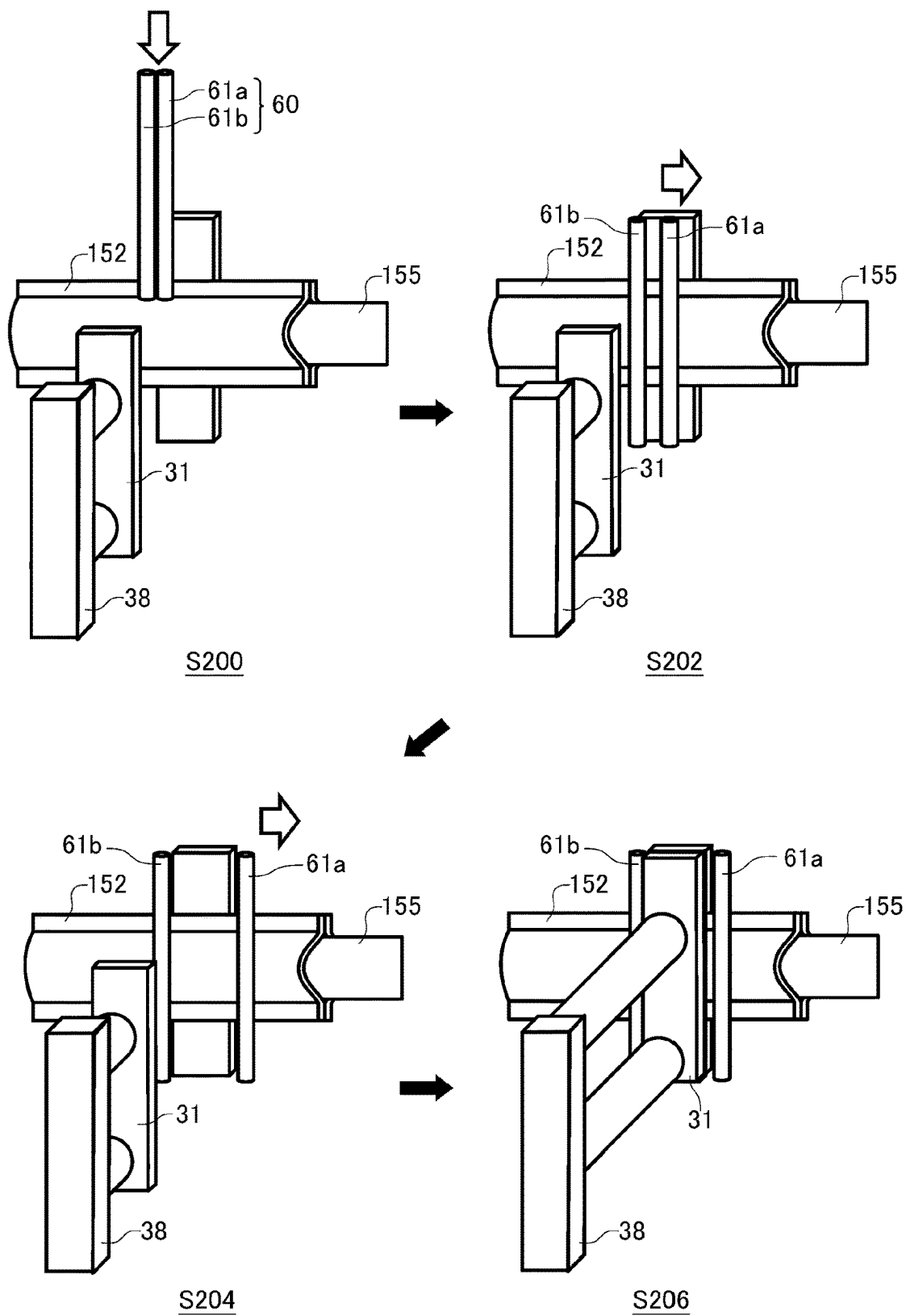


FIG. 7A

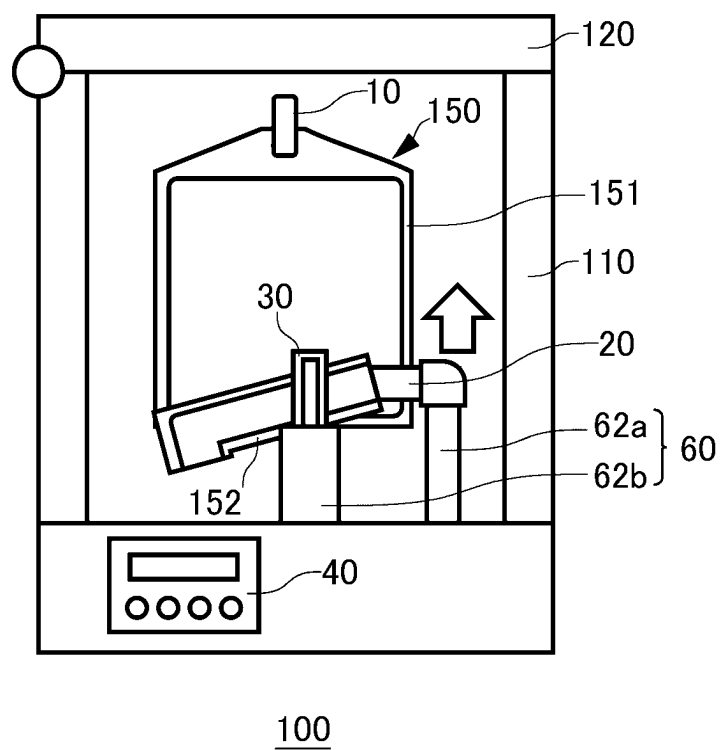


FIG. 7B

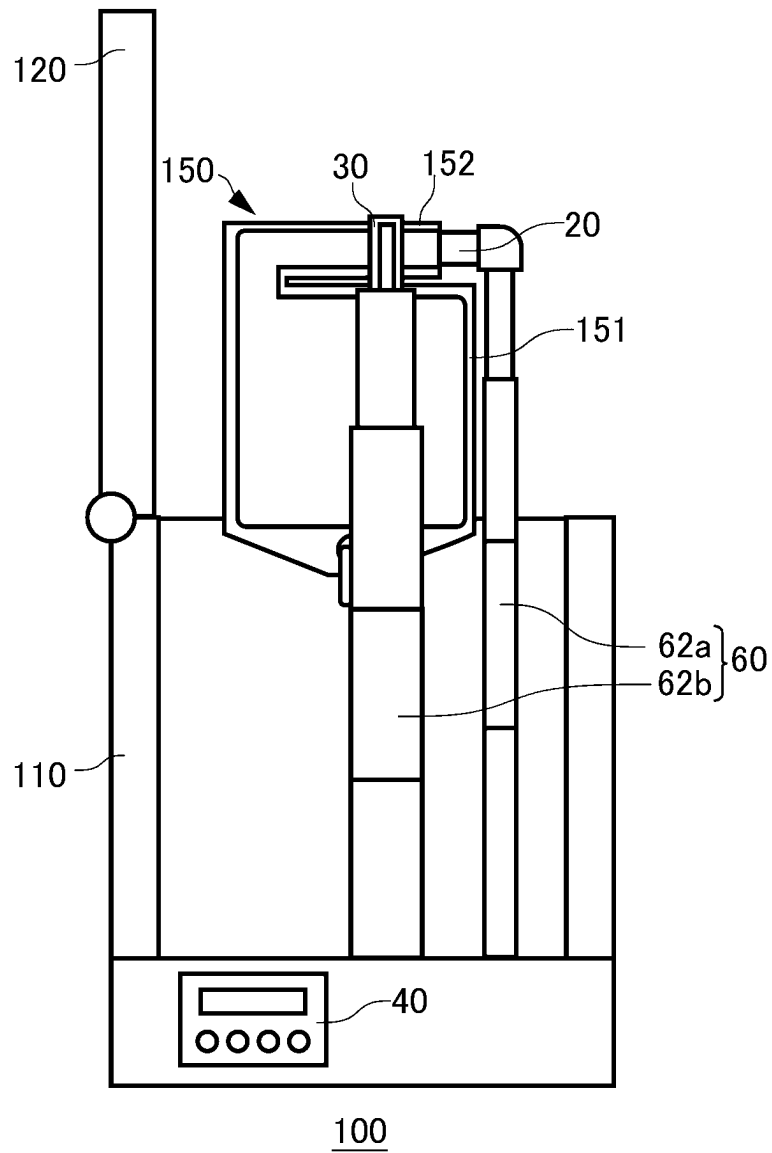


FIG. 7C

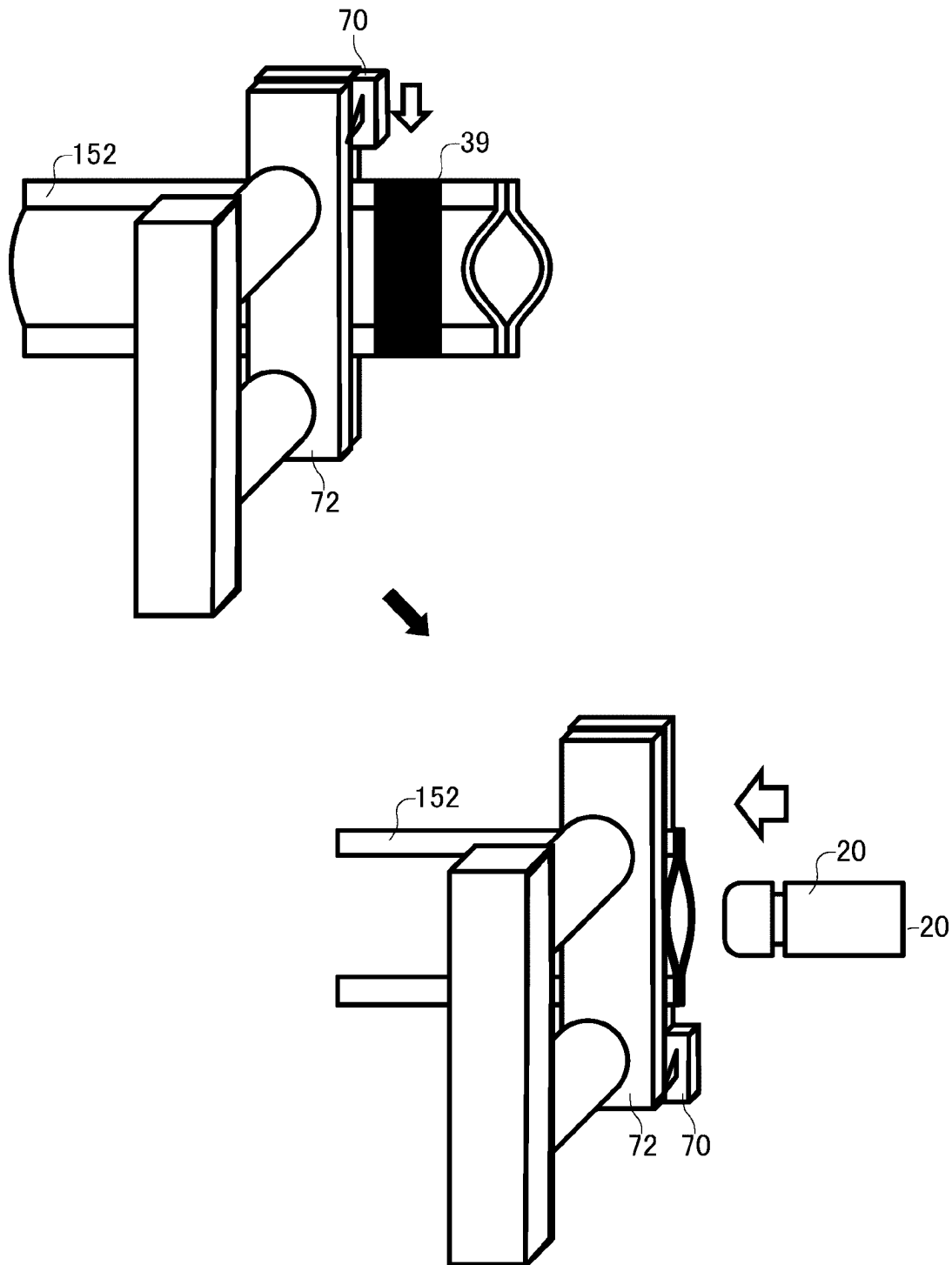


FIG. 8

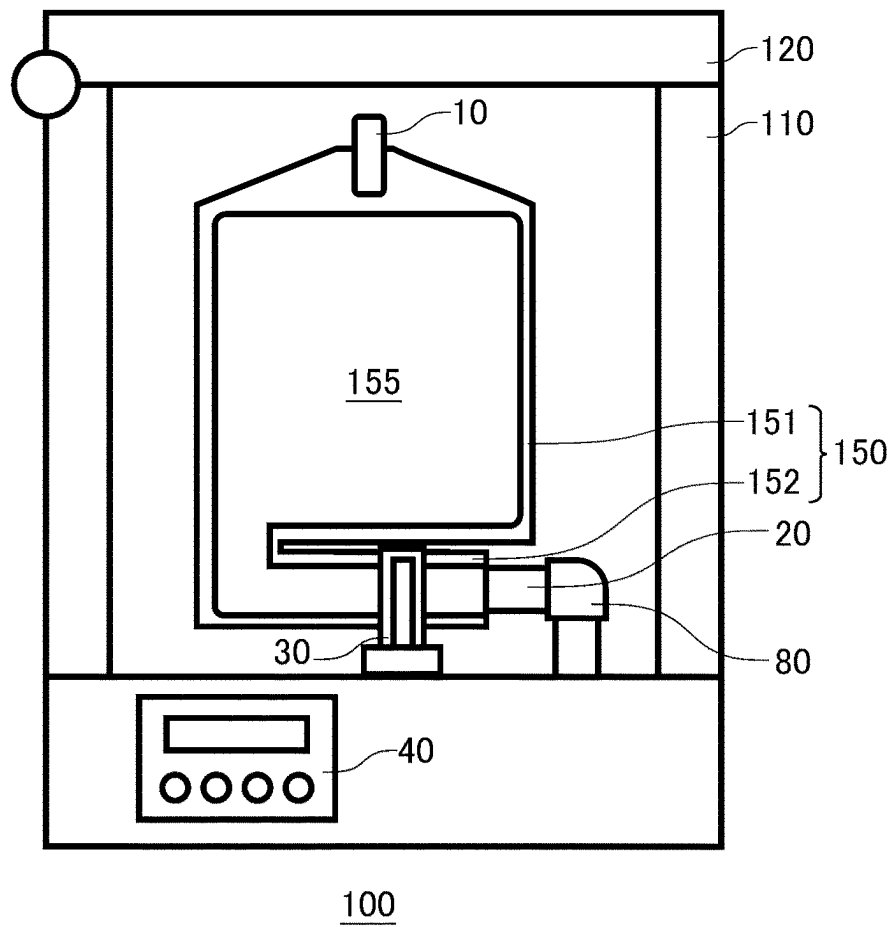


FIG. 9A

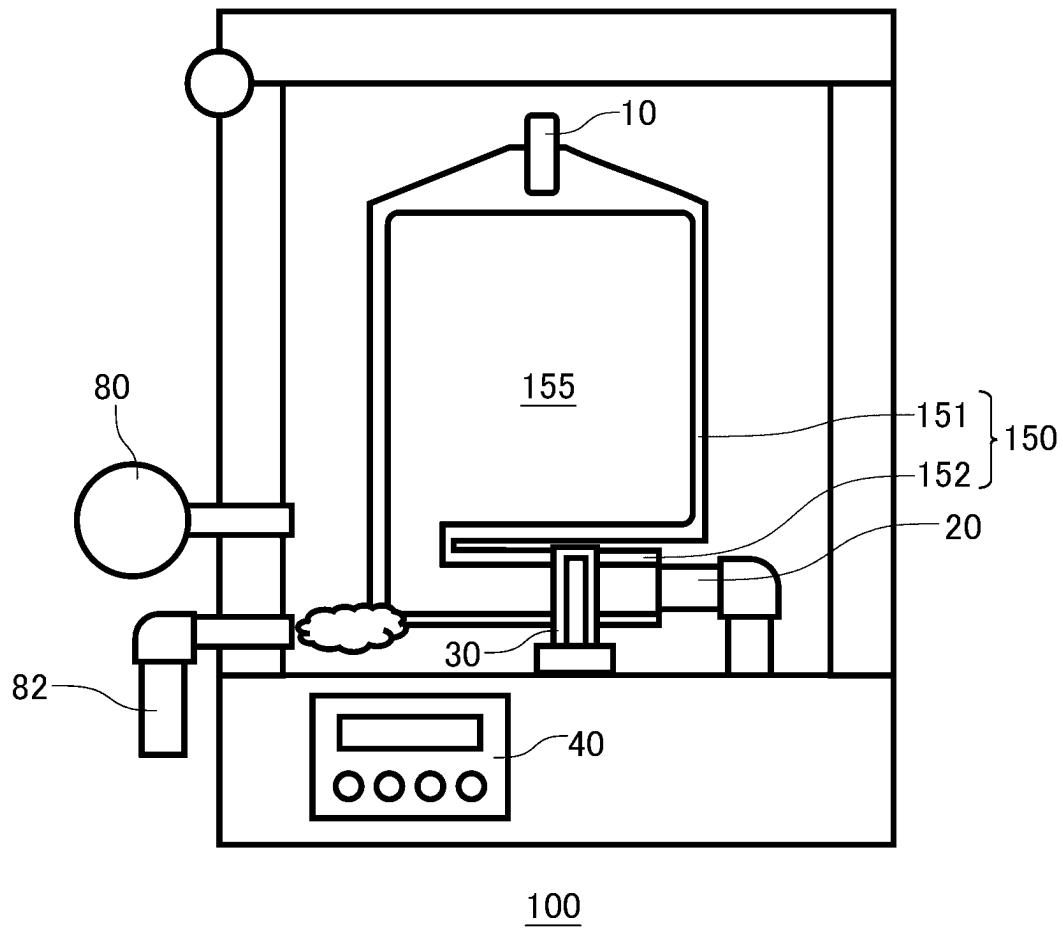


FIG. 9B

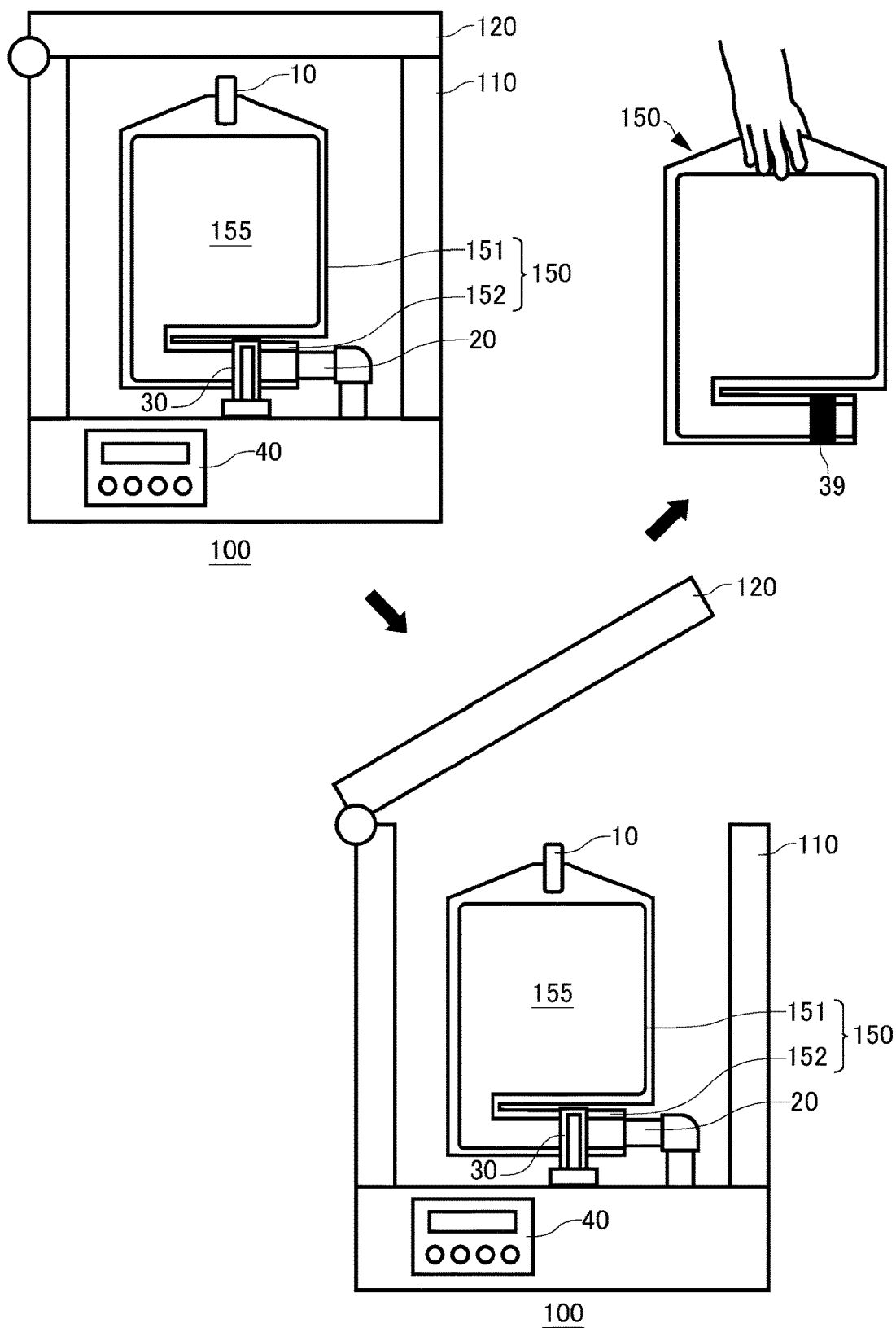


FIG. 10

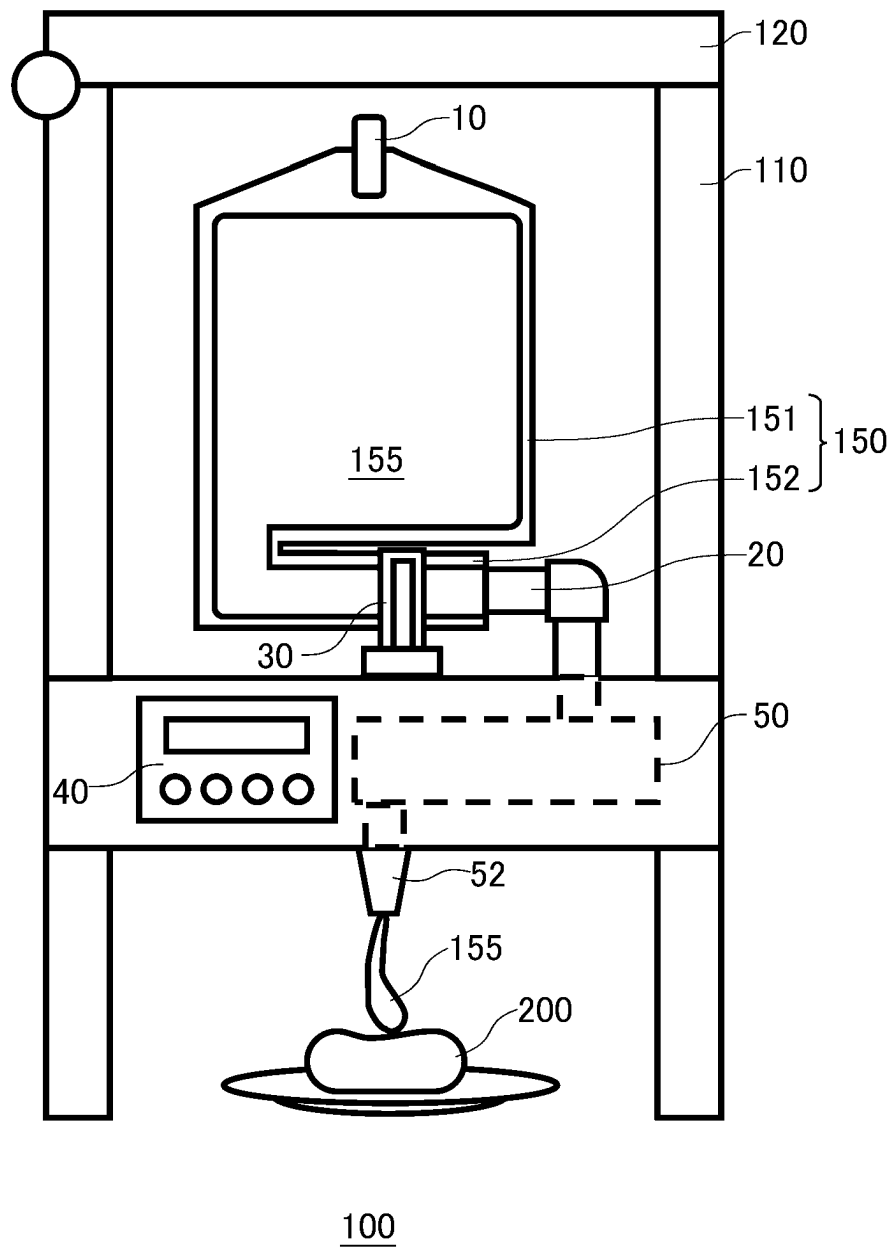


FIG. 11

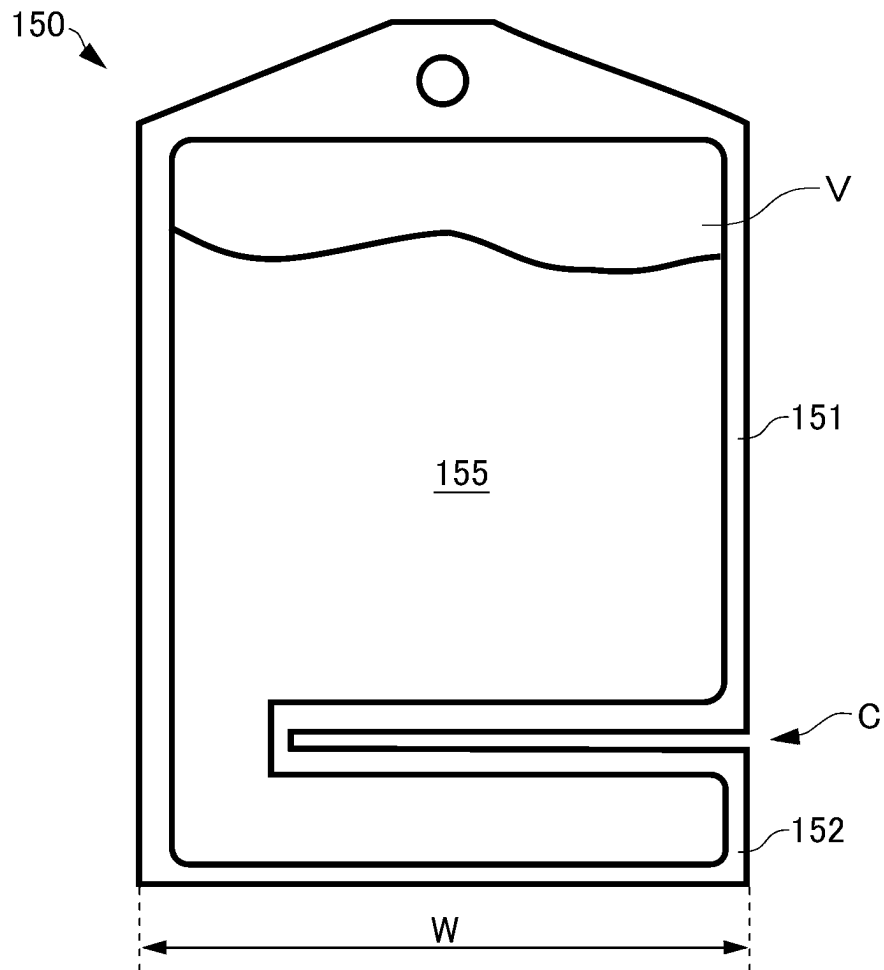


FIG. 12

1

CLOSURE DEVICE AND SOFT PACKAGING CONTAINER

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a National Stage Entry of International Application NO. PCT/JP2021/005777 filed in WO on Feb. 16, 2021, which claims priority to Japanese Patent Application NO. 2020-034292 filed in JP on Feb. 28, 2020, the contents of each of which are hereby incorporated by reference in their entirety.

BACKGROUND

1. Technical Field

The present invention relates to a closure device and a soft packaging container.

2. Related Art

Up to now, a pouch for accommodating contents has been known (for example, see Patent Document 1).

Patent Document 1: Japanese Patent Application Publication No. 2019-172355

Technical Problem

Up to now, control on timing for opening and closing a pouch has been difficult in some cases.

GENERAL DISCLOSURE

According to a first aspect of the present invention, there is provided a closure device for closing a soft packaging container, the closure device including: a holding section configured to hold a soft packaging container; a dispense section configured to dispense contents of the soft packaging container; a closure section configured to close the soft packaging container; and a control section configured to control timing for closing the soft packaging container by the closure section.

The control section may include an acquisition section configured to acquire time information related to timing for closing the soft packaging container. The control section may close the soft packaging container according to the time information acquired by the acquisition section.

The control section may include a sensing section configured to sense a content amount of the soft packaging container.

The closure section may close the soft packaging container when a content amount of the soft packaging container becomes lower than a predetermined amount.

The closure section may have a thermal welding mechanism configured to close the soft packaging container by thermal welding.

The closure section may have a closure member applying mechanism configured to apply a closure member to the soft packaging container to be closed.

The closure section may have a folding mechanism configured to fold the soft packaging container.

The closure section may have a restraint section applying mechanism configured to restraint a folded section applied by the folding mechanism.

2

The closure section may have a pinching mechanism configured to pinch the soft packaging container to be closed.

The closure device may have a failure avoiding section configured to avoid failure upon closure by moving the contents in a closure region of the soft packaging container.

The failure avoiding section may have a squeezing mechanism configured to squeeze the soft packaging container. The squeezing mechanism may squeeze contents of the soft packaging container to move the contents in the closure region.

The failure avoiding section may have an orientation change section configured to change an orientation of the soft packaging container. The orientation change section may move the contents from the closure region by a change in an orientation of the soft packaging container.

The closure device may have a cutting section configured to unseal the closed soft packaging container by cutting.

The closure device may have a closure detection section configured to detect closure of the soft packaging container.

The closure device may have an accommodation section configured to accommodate the soft packaging container. The accommodation section may be opened after closure of the soft packaging container.

The dispense section may include an ejection device configured to eject the contents of the soft packaging container.

According to a second aspect of the present invention, there is provided a soft packaging container for accommodating contents, the soft packaging container including: a main region for accommodating the contents; and an extension region provided by being extended from the main region.

The extension region may be provided along an edge of the soft packaging container. The main region and the extension region may form a C-shaped structure.

The extension region may have a length that is 50% or more of a width of the soft packaging container.

The summary clause does not necessarily describe all necessary features of the embodiments of the present invention. The present invention may be a sub-combination of the features described above.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A illustrates an example of a configuration of a closure device **100**.

FIG. 1B illustrates an example of an embodiment of the closure device **100**.

FIG. 2 illustrates an outline of the configuration of the closure device **100**.

FIG. 3 illustrates an example of the configuration of the closure device **100** including a sensing section **44**.

FIG. 4A illustrates an example of a configuration of a closure section **30**.

FIG. 4B illustrates an example of the configuration of the closure section **30**.

FIG. 4C illustrates an example of the configuration of the closure section **30**.

FIG. 5A illustrates an example of a restraint section **36**.

FIG. 5B illustrates an example of the restraint section **36**.

FIG. 5C illustrates an example of the restraint section **36**.

FIG. 6 illustrates an example of the closure section **30** having a pinching mechanism **37**.

FIG. 7A illustrates an example of a configuration of a failure avoiding section **60**.

3

FIG. 7B illustrates an example of the configuration of the failure avoiding section 60.

FIG. 7C illustrates an example of the configuration of the failure avoiding section 60.

FIG. 8 illustrates an example of a cutting section 70.

FIG. 9A illustrates an example of the closure device 100 including a closure detection section 80.

FIG. 9B illustrates an example of the closure device 100 including the closure detection section 80.

FIG. 10 illustrates an example of the configuration of the closure device 100 including a locking mechanism.

FIG. 11 illustrates an example of the configuration of the closure device 100.

FIG. 12 illustrates an example of a shape of a soft packaging container 150.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

Hereinafter, the present invention will be described through embodiments of the invention, but the following embodiments do not limit the invention according to the claims. In addition, not all combinations of the features described in the embodiments necessarily have to be essential to solving means of the invention.

FIG. 1A illustrates an example of a configuration of a closure device 100. The closure device 100 includes an accommodation section 110 and a lid section 120. In addition, the closure device 100 includes a holding section 10, a dispense section 20, a closure section 30, and a control section 40. The present example illustrates states before and after the closure device 100 accommodates a soft packaging container 150.

The closure device 100 accommodates the soft packaging container 150 and dispenses contents 155 from the unsealed soft packaging container 150. The closure device 100 closes the unsealed soft packaging container 150. The closure device 100 accommodates the soft packaging container 150 in the accommodation section 110. The lid section 120 is a lid of the accommodation section 110 accommodating the soft packaging container 150. In the closure device 100, the soft packaging container 150 can be taken out and put in by opening and closing the lid section 120.

The holding section 10 is configured to hold the soft packaging container 150. For example, the holding section 10 has a hook shape to be hooked at an opening provided in the soft packaging container 150. The holding section 10 may hold the soft packaging container 150 by nipping. The holding section 10 of the present example holds only an upper end of the soft packaging container 150 but may hold an entirety of the soft packaging container 150.

The dispense section 20 is configured to dispense the contents 155 from the soft packaging container 150. The dispense section 20 is inserted into the opening of the soft packaging container 150. For example, the dispense section 20 ejects the contents 155 from an external nozzle of the closure device 100.

The closure section 30 is configured to close the unsealed soft packaging container 150. Closure refers to closing of an outlet in a state where the contents 155 cannot be dispensed from the soft packaging container 150. That is, the closure section 30 may change the soft packaging container 150 from an unsealed state into a sealed state, or may change the state into a state where the contents 155 cannot be dispensed by folding the opening of the soft packaging container 150 or the like.

4

The control section 40 is configured to control timing for closing the soft packaging container 150 by the closure section 30. For example, the control section 40 controls closure timing of the soft packaging container 150 according to a content amount, expiry date, or the like of the contents 155. In an example, when an amount of the contents 155 of the soft packaging container 150 becomes lower than a predetermined amount, the control section 40 causes the closure section 30 to close the soft packaging container 150.

The soft packaging container 150 accommodates the contents 155. The soft packaging container 150 includes a main region 151 and an extension region 152. A capacity of the main region 151 is larger than a capacity of the extension region 152. The extension region 152 is provided by being extended from the main region 151. The extension region 152 is connected to the dispense section 20. An opening for dispensing the contents 155 is provided in the extension region 152, and is closed by the closure section 30. The soft packaging container 150 will be described below.

FIG. 1B illustrates an example of an embodiment of the closure device 100. The closure device 100 of the present example applies Japanese horseradish corresponding to the contents 155 to vinegared rice corresponding to an object 200. The closure device 100 of the present example includes a nozzle 52.

For example, the closure section 30 closes the soft packaging container 150 after the end of business hours. In addition, the closure section 30 closes the soft packaging container 150 after the use of the contents 155. In this manner, contamination of the closure device 100 or the like due to leakage of the contents 155 is prevented upon replacement of the soft packaging container 150.

The nozzle 52 is configured to extract the contents 155 towards the object 200. The nozzle 52 may have a shape according to a use. The nozzle 52 of the present example has a shape appropriate to extract Japanese horseradish corresponding to the contents 155 to vinegared rice corresponding to the object 200.

FIG. 2 illustrates an outline of the configuration of the closure device 100. The control section 40 includes an acquisition section 41 and a closure instruction section 45.

The acquisition section 41 is configured to acquire information required to decide the closure timing. The acquisition section 41 has a time information output section 42, an information output section 43, and a sensing section 44.

The time information output section 42 is configured to output time information related to timing for closing the soft packaging container 150. For example, the time information output section 42 has a timer, and outputs elapsed time since the soft packaging container 150 is unsealed to the closure instruction section 45. In addition, the time information output section 42 may have clock time acquisition means such as a clock, and output a current time to the closure instruction section 45.

The information output section 43 is configured to acquire information required to decide the closure timing of the soft packaging container 150. For example, the information output section 43 outputs an expiry date of the soft packaging container 150 to the closure instruction section 45. The information output section 43 may directly acquire the expiry date from the soft packaging container 150 accommodated in the closure device 100 or may acquire information input by a user or the like.

The sensing section 44 is configured to sense a content amount of the soft packaging container 150. The sensing section 44 may sense a total capacity of the soft packaging container 150 before use or may sense a remaining amount

5

of the contents 155. For example, the sensing section 44 senses the content amount of the soft packaging container 150 based on a weight of the soft packaging container 150.

The closure instruction section 45 is configured to decide whether the soft packaging container 150 is to be closed according to input information. For example, the closure instruction section 45 closes the soft packaging container 150 according to at least one of the time information, the expiry date, or the content amount acquired by the acquisition section 41. When the soft packaging container 150 is to be closed, the closure instruction section 45 inputs a closure command to the closure section 30.

FIG. 3 illustrates an example of the configuration of the closure device 100 including the sensing section 44. The closure device 100 of the present example is different from the closure device 100 of FIG. 1A in that the sensing section 44 is included. In the present example, an aspect different from the case of FIG. 1A will be described in particular.

The sensing section 44 senses a remaining amount of the contents 155 of the soft packaging container 150. The sensing section 44 of the present example is a weight sensor configured to sense a remaining amount of the contents 155 by measuring a weight of the soft packaging container 150. The sensing section 44 is coupled to the holding section 10 and measures the weight of the soft packaging container 150.

The control section 40 may take into account the weight of the soft packaging container 150 itself in the determination of the closure timing or does not need to take the weight into account. For example, the control section 40 closes the soft packaging container 150 when the sensed weight becomes lower than a predetermined value without taking into account the weight of the soft packaging container 150 itself. In addition, the control section 40 may acquire information from a two-dimensional code of the soft packaging container 150 and detect the remaining amount of the contents 155 by taking into account the weight of the soft packaging container 150 itself.

FIG. 4A illustrates an example of a configuration of the closure section 30. The closure section 30 of the present example has a thermal welding mechanism 31 and a drive section 38.

The thermal welding mechanism 31 is configured to close the soft packaging container 150 by thermal welding. For example, the thermal welding mechanism 31 is a heat sealer for closing the soft packaging container 150. The thermal welding mechanism 31 of the present example is arranged so as to face a closure position of the extension region 152. In order to prevent thermal welding failure, the thermal welding mechanism 31 may be arranged in a region where it is difficult for the contents 155 to stay. The closure section 30 of the present example can prevent leakage of the contents 155 by closing an opening of the extension region 152 by the thermal welding mechanism 31.

The drive section 38 is configured to drive the thermal welding mechanism 31 so as to nip the extension region 152. The drive section 38 of the present example nips the extension region 152 by moving a position of the thermal welding mechanism 31 by an air cylinder. The drive section 38 drives the thermal welding mechanism 31 at timing at which the control section 40 instructs closure of the soft packaging container 150.

The closure section 30 of the present example closes the extension region 152 in a state where the soft packaging container 150 is removed from the dispense section 20, but may close the extension region 152 in a state where the dispense section 20 is inserted into the extension region 152.

6

In this case, the thermal welding mechanism 31 is arranged so as to avoid the dispense section 20, and performs thermal welding of the extension region 152.

FIG. 4B illustrates an example of the configuration of the closure section 30. The closure section 30 of the present example has a closure member applying mechanism 32 and a closure member 33.

The closure member applying mechanism 32 applies the closure member 33 to the soft packaging container 150 to be closed. For example, the closure member applying mechanism 32 nips the extension region 152 by the closure member 33 to close the soft packaging container 150. For example, the closure member applying mechanism 32 is a robot hand that can move in a state where the closure member 33 is grabbed at will or release the closure member 33.

The closure member 33 is applied to the soft packaging container 150 to close the extension region 152. The closure member 33 of the present example is a clip but is not limited to this. The closure member 33 may be a rubber band, an adhesive tape, or the like for closing the extension region 152.

The closure member applying mechanism 32 of the present example nips the extension region 152 with the clip by vertically moving the closure member 33. After the closure member 33 is attached to the extension region 152, the closure member applying mechanism 32 releases the closure member 33. Therefore, the soft packaging container 150 can be removed to the outside of the closure device 100 in a state where the closure member 33 is mounted.

FIG. 4C illustrates an example of the configuration of the closure section 30. The closure section 30 of the present example has a folding mechanism 34.

The folding mechanism 34 is configured to fold the soft packaging container 150. The closure section 30 of the present example has two bar-shaped folding mechanism 34a and folding mechanism 34b. For example, the folding mechanism 34a and the folding mechanism 34b are robot fingers that can be manipulated at will.

A restraint section applying mechanism 35 is configured to apply a restraint section 36 which will be described below to the extension region 152 so as to prevent the folded extension region 152 from returning to its original state. For example, the restraint section applying mechanism 35 closes the extension region 152 by thermal welding.

In step S100, the folding mechanism 34a and the folding mechanism 34b nip the extension region 152. Before step S100, the dispense section 20 may be extracted from the soft packaging container 150, or after step S100, the dispense section 20 may be extracted from the soft packaging container 150.

In step S102, the extension region 152 is turned down by the folding mechanism 34b with the folding mechanism 34a set as a supporting point. In step S104, the extension region 152 is maintained in a folded state by the folding mechanism 34b.

In step S106, while the folding of the extension region 152 is maintained by the folding mechanism 34b, the folding mechanism 34a set as the supporting point is extracted. In step S108, the restraint section applying mechanism 35 is pressed at a closure position of the extension region 152. In step S110, the folding mechanism 34b is extracted. In this manner, the folding mechanism 34 realizes the folding and the thermal welding of the extension region 152.

FIG. 5A illustrates an example of the restraint section 36. The restraint section 36 is applied to the folded extension region 152 by the restraint section applying mechanism 35.

The restraint section **36** is configured to restrain a folded section applied by the folding mechanism **34**. Since the restraint section **36** of the present example is a clip, the closed soft packaging container **150** is easily unsealed.

FIG. **5B** illustrates an example of the restraint section **36**. The restraint section **36** is a region where the extension region **152** has been subjected to thermal welding by the restraint section applying mechanism **35**. The extension region **152** of the present example is closed by folding, and maintains the folded state by thermal welding. That is, the restraint section **36** of the present example is obtained by thermal welding of mutual external surfaces of the extension region **152**, and it does not need to perform thermal welding of an inner side of the extension region **152** through which the contents **155** passes. Therefore, thermal welding failure hardly occurs due to residue of the contents **155**.

FIG. **5C** illustrates an example of the restraint section **36**. The restraint section **36** is an adhesive tape. Since the restraint section **36** of the present example is an adhesive tape, the closed soft packaging container **150** is easily unsealed. In addition, the restraint section **36** of the present example can be disposed together with the spent soft packaging container **150**. Note that the restraint sections **36** illustrated in FIG. **5A** to FIG. **5C** may be mutually combined.

FIG. **6** illustrates an example of the closure section **30** having a pinching mechanism **37**. The pinching mechanism **37** is configured to pinch the soft packaging container **150** to be closed. The closure section **30** of the present example has a pair of a pinching mechanism **37a** and a pinching mechanism **37b**. The pinching mechanism **37a** has a mechanism configured to rotate about the pinching mechanism **37b**. The closure section **30** pinches the extension region **152** by arranging the extension region **152** between the pinching mechanism **37a** and the pinching mechanism **37b** and rotating the pinching mechanism **37a** for closing.

The pinching mechanism **37** may be fixed to any configuration of the closure device **100**. For this reason, a configuration may be adopted where the soft packaging container **150** cannot be removed from the closure device **100** in a closed state by the pinching mechanism **37**. When the soft packaging container **150** is removed from the closure device **100**, the soft packaging container **150** may be separated from the pinching mechanism **37**. Therefore, in the closure device **100** of the present example, the pinching mechanism **37** can be repeatedly used even when the soft packaging container **150** is replaced.

FIG. **7A** illustrates an example of a configuration of a failure avoiding section **60**. The failure avoiding section **60** of the present example has a squeezing mechanism **61**. The failure avoiding section **60** of the present example has two squeezing mechanism **61a** and squeezing mechanism **61b**.

The failure avoiding section **60** is configured to prevent residue of the contents **155** by moving the contents **155** in the closure region of the soft packaging container **150**. In this manner, the failure avoiding section **60** avoids the failure upon closure of the soft packaging container **150**.

The squeezing mechanism **61** is used to squeeze the soft packaging container **150**. The squeezing mechanism **61** is configured to move the contents **155** in the closure region by squeezing the contents **155** of the soft packaging container **150**. For example, the squeezing mechanism **61** is robot fingers for squeezing.

In step **S200**, the squeezing mechanism **61a** and the squeezing mechanism **61b** are vertically moved to be arranged at positions corresponding to the closure region of the extension region **152**. In step **S202**, in a state where the

squeezing mechanism **61b** is fixed, the squeezing mechanism **61a** is slid to an opening side of the extension region **152**. The squeezing mechanism **61b** may be pressed against the extension region **152** to restrict movement of the contents **155** from the main region **151** side to the opening side.

In step **S204**, the contents **155** in the closure region is discharged by sliding of the squeezing mechanism **61a**. In step **S206**, thermal welding is performed by pressing the thermal welding mechanism **31** against the extension region **152** between the squeezing mechanism **61a** and the squeezing mechanism **61b**. In the present example, since the contents **155** in the closure region is discharged, the failure upon closure by the contents **155** can be avoided.

FIG. **7B** illustrates an example of the configuration of the failure avoiding section **60**. The failure avoiding section **60** of the present example has an orientation change section **62**.

The orientation change section **62** is configured to change an orientation of the soft packaging container **150**. For example, the orientation change section **62** changes an orientation of the extension region **152**. The orientation change section **62** moves the contents **155** from the closure region by the change in the orientation of the soft packaging container **150**. The failure avoiding section **60** of the present example includes an orientation change section **62a** and an orientation change section **62b**.

The orientation change section **62a** is connected to the dispense section **20** and changes a position of the dispense section **20**. The orientation change section **62a** of the present example is configured to change a position of the opening of the extension region **152** by changing a position of the dispense section **20**. The orientation change section **62a** may change the position of the dispense section **20** in not only the vertical direction but also a horizontal direction.

The orientation change section **62b** is connected to the closure section **30** and changes a position of the closure section **30**. The orientation change section **62b** may change the position of the closure section **30** according to control of the orientation change section **62a**. For example, the orientation change section **62b** may change the position of the closure section **30** in not only the vertical direction but also the horizontal direction according to a position of the extension region **152**.

The orientation change section **62a** of the present example raises the position of the opening of the extension region **152** by extending upwards. In this manner, since the contents **155** in the closure region of the extension region **152** flows onto an opposite side of the opening of The extension region **152**, the failure upon closure of the soft packaging container **150** can be avoided. Note that the orientation change section **62b** of the present example may extend upwards or does not need to extend.

FIG. **7C** illustrates an example of the configuration of the failure avoiding section **60**. Similarly as in FIG. **7B**, the failure avoiding section **60** of the present example has the orientation change section **62**. In the present example, a change degree of the orientation of the soft packaging container **150** is different from the embodiment of FIG. **7B**.

The orientation change section **62a** and the orientation change section **62b** change the orientation of the soft packaging container **150** upside down by extending upwards. In this manner, since the contents **155** in the closure region of the extension region **152** flows to the main region **151**, the failure upon closure of the soft packaging container **150** can be avoided.

The closure device **100** of the present example arranges the soft packaging container **150** outside the accommodation section **110** by extending the orientation change section **62**.

For this reason, the soft packaging container **150** is closed, and the soft packaging container **150** can be immediately removed. Note that the closure device **100** may accommodate the soft packaging container **150** inside the accommodation section **110** even in a vertically inverted state.

FIG. **8** illustrates an example of a cutting section **70**. The closure device **100** of the present example includes the cutting section **70** and a fixed section **72**. The extension region **152** has a thermal welding section **39** having been subjected to thermal welding by the thermal welding mechanism **31**.

The cutting section **70** is configured to unseal the soft packaging container **150** by cutting the closed soft packaging container **150**. The cutting section **70** of the present example unseals the soft packaging container **150** by cutting the extension region **152**. In this manner, after the soft packaging container **150** is accommodated in the closure device **100** while being still sealed, the closure device **100** can unseal the soft packaging container **150**. The cutting section **70** may unseal the soft packaging container **150** in a state where the orientation of the soft packaging container **150** is fixed such that the contents **155** is not to be leaked.

The fixed section **72** nips and fixes the extension region **152** in order to cut the extension region **152**. The fixed section **72** of the present example closes a path of the contents **155** in order to prevent the leakage of the contents **155** when unsealed. It is noted however that the fixed section **72** does not need to close the path of the contents **155** in such an orientation that the contents **155** is not to be leaked.

Note that the fixed section **72** may have a thermal welding function of the extension region **152**. In this case, the closure device **100** can repeat closure and unsealing of the soft packaging container **150**. Since the extension region **152** is shortened when the closure and the unsealing of the soft packaging container **150** are repeated, positions of the closure section **30** and the cutting section **70** may be adjusted according to a length of the extension region **152**.

FIG. **9A** illustrates an example of the closure device **100** including a closure detection section **80**. The closure device **100** of the present example includes a flowmeter as the closure detection section **80** which is configured to measure a flow rate of the contents **155**.

The closure detection section **80** is configured to detect closure of the soft packaging container **150** by measuring a flow rate of the contents **155**. For example, the closure detection section **80** determines that the soft packaging container **150** is closed when the flow rate of the contents **155** is lower than a predetermined flow rate. The closure detection section **80** of the present example is provided in the dispense section **20**.

According to another embodiment, the closure detection section **80** may calculate the flow rate of the contents **155** from a change in the weight of the soft packaging container **150** instead of measuring the flow rate of the contents **155**. In this case, the closure detection section **80** determines that the soft packaging container **150** is closed when a change amount of the weight of the soft packaging container **150** is lower than a predetermined value.

FIG. **9B** illustrates an example of the closure device **100** including the closure detection section **80**. The closure device **100** of the present example includes a manometer as the closure detection section **80** which is configured to measure a pressure inside the accommodation section **110**.

The closure detection section **80** is configured to detect closure of the soft packaging container **150** by measuring the pressure inside the accommodation section **110**. For example, in a case where the closure of the soft packaging

container **150** is satisfactory, when a pressure source **82** is actuated, a pressure is increased. On the other hand, in the case of closure failure, since the soft packaging container **150** collapses, a pressure increase speed is decreased.

FIG. **10** illustrates an example of the configuration of the closure device **100** including a locking mechanism. The closure device **100** of the present example is configured to prevent unintended opening and closing of the accommodation section **110** by locking the lid section **120**. For example, the closure device **100** opens the accommodation section **110** after the soft packaging container **150** is closed. In this manner, the closure device **100** prevents contamination by the contents **155** upon opening and closing of the accommodation section **110** due to carelessness of a user or the like. In addition, the closure device **100** may protect the user from high heat by locking the lid section **120** upon thermal welding.

FIG. **11** illustrates an example of the configuration of the closure device **100**. The closure device **100** of the present example includes an ejection device **50**.

The ejection device **50** is configured to eject the contents **155** of the soft packaging container **150**. The ejection device **50** of the present example is coupled to the dispense section **20** and ejects the contents **155** dispensed by the dispense section **20** from the nozzle **52**. For example, the ejection device **50** ejects the contents **155** using a pump. The ejection device **50** may decide an amount of the contents **155** to be ejected each time, a flow rate of the contents **155**, and the like.

FIG. **12** illustrates an example of a shape of the soft packaging container **150**. The soft packaging container **150** of the present example has the main region **151** and the extension region **152**.

The main region **151** and the extension region **152** are separated from each other by a slit **C**. A slit stop section may be provided at an edge of the slit **C** so as not to generate a fissure when the slit **C** is formed. For example, the generation of the fissure can be prevented by providing an opening at the edge of the slit **C** in advance.

The extension region **152** is provided along an edge of the soft packaging container **150**. In this manner, the extension region **152** can freely adjust the position according to the position of the dispense section **20**. In addition, folding of the extension region **152** is facilitated. Then, the position of the opening of the extension region **152** can be easily freely adjusted according to the orientation of the soft packaging container **150**.

The main region **151** and the extension region **152** form a C-shaped structure. Since the extension region **152** is provided, the soft packaging container **150** of the present example more easily avoids thermal welding failure by the thermal welding mechanism **31** as compared with a case where the extension region **152** is not provided.

A spatial region **V** is a region that is not occupied by the contents **155** inside the soft packaging container **150**. A volume of the spatial region **V** is equal to or more than a volume of the extension region **152**. In other words, the volume of the contents **155** in a still sealed state may be smaller than a spatial volume of the main region **151**. In this manner, the dispense section **20** is easily mounted since the extension region **152** can be emptied by accommodating the contents **155** in only the main region **151**. In addition, when the soft packaging container **150** is to be closed immediately after being unsealed, since the extension region **152** is not yet filled, closure failure hardly occurs. Furthermore, the soft

11

packaging container **150** has a margin of the spatial region **V** in a still sealed state, handling such as folding is facilitated.

The extension region **152** has a length that is 50% or more of a width **W** of the soft packaging container **150**. The extension region **152** may have a length that is 75% or more, or may have a length that is 90% or more, of the width **W** of the soft packaging container **150**. In the present example, the length of the extension region **152** is the same as the width **W** of the soft packaging container **150**.

The length of the extension region **152** may be longer, or may be shorter, than the width **W** of the soft packaging container **150**. The length of the extension region **152** may be decided by taking into account an insertion length of the dispense section **20** into the extension region **152**, a length required for closure by the closure section **30**, a flow path length from the main region **151** to the extension region **152**, or the like. For example, the length of the extension region **152** is at least 40 mm or more. In addition, the length of the extension region **152** is 40 mm or more and may be equal to or less than the width **W** of the soft packaging container **150**.

While the embodiments of the present invention have been described, the technical scope of the present invention is not limited to the above described embodiments. It is apparent to persons skilled in the art that various alterations and improvements can be added to the above-described embodiments. It is also apparent from the scope of the claims that the embodiments added with such alterations or improvements can be included in the technical scope of the present invention.

The operations, procedures, steps, and stages of each process performed by an apparatus, system, program, and method shown in the claims, embodiments, or diagrams can be performed in any order as long as the order is not indicated by "prior to," "before," or the like and as long as the output from a previous process is not used in a later process. Even if the process flow is described using phrases such as "first" or "next" in the claims, embodiments, or diagrams, it does not necessarily mean that the process must be performed in this order.

EXPLANATION OF REFERENCES

10: holding section, **20**: dispense section, **30**: closure section, **31**: thermal welding mechanism, **32**: closure member applying mechanism, **33**: closure member, **34**: folding mechanism, **35**: restraint section applying mechanism, **36**: restraint section, **37**: pinching mechanism, **38**: drive section, **39**: thermal welding section, **40**: control section, **41**: acquisition section, **42**: time information output section, **43**: information output section, **44**: sensing section, **45**: closure instruction section, **50**: ejection device, **52**: nozzle, **60**: failure avoiding section, **61**: squeezing mechanism, **62**: orientation change section, **70**: cutting section, **72**: fixed section, **80**: closure detection section, **82**: pressure source, **100**: closure device, **110**: accommodation section, **120**: lid section, **150**: soft packaging container, **151**: main region, **152**: extension region, **155**: contents, **200**: object

What is claimed is:

1. A closure device for closing a soft packaging container, the closure device comprising:

a holding section configured to hold a soft packaging container;

a dispense section configured to dispense contents of the soft packaging container;

12

a closure section configured to close the soft packaging container; and

a control section configured to control timing for closing the soft packaging container by the closure section, wherein

the control section includes an acquisition section configured to acquire time information related to timing for closing the soft packaging container, and

the control section is configured to close the soft packaging container according to the time information acquired by the acquisition section.

2. The closure device according to claim **1**, wherein the control section includes a sensing section configured to sense a content amount of the soft packaging container.

3. The closure device according to claim **1**, wherein the closure section is configured to close the soft packaging container when a content amount of the soft packaging container becomes lower than a predetermined amount.

4. The closure device according to claim **1**, wherein the closure section has a closure member applying mechanism configured to apply a closure member to the soft packaging container to be closed.

5. The closure device according to claim **1**, wherein the closure section has a folding mechanism configured to fold the soft packaging container.

6. The closure device according to claim **5**, wherein the closure section has a restraint section applying mechanism configured to restrain a folded section applied by the folding mechanism.

7. The closure device according to claim **1**, comprising: a failure avoiding section configured to avoid failure upon closure by moving contents in a closure region of the soft packaging container.

8. The closure device according to claim **7**, wherein the failure avoiding section has a squeezing mechanism configured to squeeze the soft packaging container, and the squeezing mechanism is configured to squeeze contents of the soft packaging container to move the contents in the closure region.

9. The closure device according to claim **7**, wherein the failure avoiding section has an orientation change section configured to change an orientation of the soft packaging container, and

the orientation change section is configured to move the contents from the closure region by a change in an orientation of the soft packaging container.

10. The closure device according to claim **1**, comprising: a cutting section configured to cut the closed soft packaging container to be unsealed.

11. The closure device according to claim **1**, comprising: an accommodation section configured to accommodate the soft packaging container, wherein the accommodation section is opened after closure of the soft packaging container.

12. The closure device according to claim **1**, wherein the dispense section includes an ejection device configured to eject the contents of the soft packaging container.

13. The closure device according to claim **1**, wherein the control section includes a sensing section configured to sense a content amount of the soft packaging container.

14. A closure device for closing a soft packaging container, the closure device comprising:

13

a holding section configured to hold a soft packaging container;
 a dispense section configured to dispense contents of the soft packaging container;
 a closure section configured to close the soft packaging container; and
 a control section configured to control timing for closing the soft packaging container by the closure section, wherein
 the closure section has a thermal welding mechanism configured to close the soft packaging container by thermal welding.
15. A closure device for closing a soft packaging container, the closure device comprising:
 a holding section configured to hold a soft packaging container;
 a dispense section configured to dispense contents of the soft packaging container;
 a closure section configured to close the soft packaging container; and

14

a control section configured to control timing for closing the soft packaging container by the closure section, wherein
 the closure section has a pinching mechanism configured to pinch the soft packaging container to be closed.
16. A closure device for closing a soft packaging container, the closure device comprising:
 a holding section configured to hold a soft packaging container;
 a dispense section configured to dispense contents of the soft packaging container;
 a closure section configured to close the soft packaging container; and
 a control section configured to control timing for closing the soft packaging container by the closure section; and
 a closure detection section configured to detect closure of the soft packaging container.

* * * * *